

Edit

# A.E.G.I.S. Personal AI Platform

## Comprehensive System Architecture, Functional Specification, Development Roadmap, and Future Autonomy Framework

**Project Name:** A.E.G.I.S.

**Assistant Identity:** MAX

**Expansion:** Autonomous Electronic Guardian Intelligence System

**Initial Objective:** Build a private, personalized, multimodal AI assistant with persistent memory, environmental awareness, tool use, autonomous task execution, and safe control of connected digital systems.

**Long-Term Objective:** Evolve A.E.G.I.S. into a distributed AI platform capable of operating across computers, mobile devices, vehicles, robotics, sensors, wearables, and other physical systems, with a future multi-user commercial/cloud offering.

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## 1. Executive Summary

A.E.G.I.S. is envisioned as a **personal AI operating platform**, rather than a conventional chatbot or voice assistant.

The system is inspired conceptually by the capabilities represented by fictional systems such as JARVIS, FRIDAY, and EDITH, but the engineering objective is to build an original architecture based on currently available AI, software, networking, robotics, perception, and embedded technologies.

The central idea is:

**One persistent intelligence, multiple interfaces, multiple devices, multiple environments, and controlled autonomy.**

The system will initially operate as a software-first personal AI. It will use the user's PC and Mac as development and compute environments while gradually integrating voice, vision, memory, operating-system control, web access, applications, and autonomous workflows.

Later versions will extend A.E.G.I.S. into:

- smartphones
- cameras and sensors
- embedded systems
- vehicles
- robotics
- smart glasses and wearables
- physical autonomous systems
- distributed edge devices

A.E.G.I.S. is divided into five architectural layers:

A — Autonomous

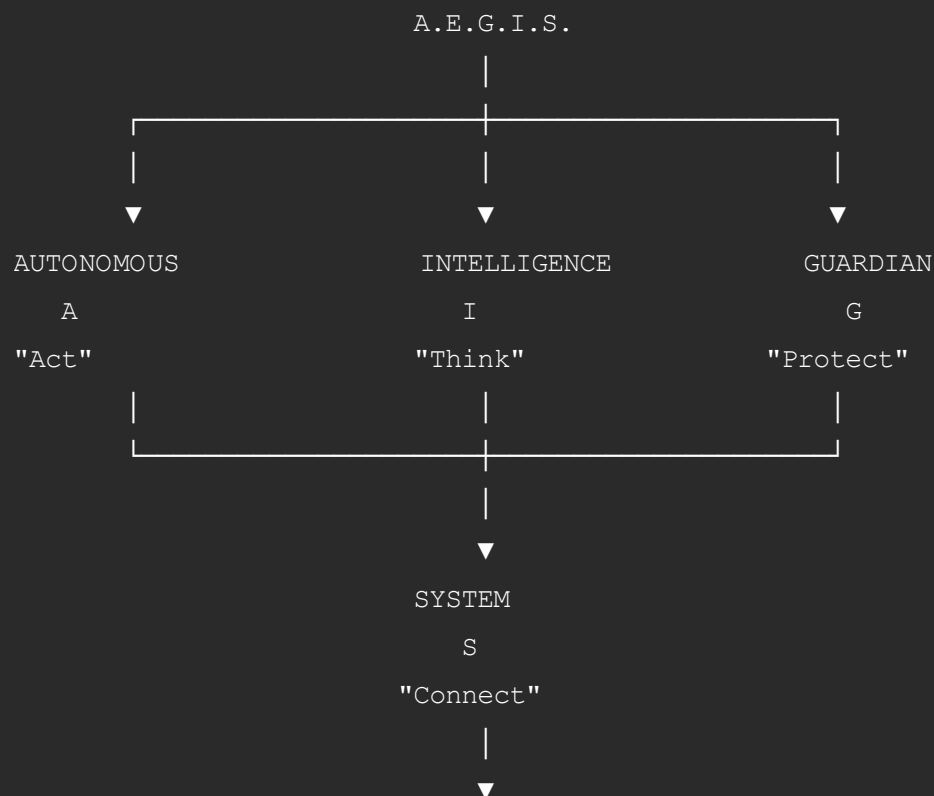
E — Electronic

G — Guardian

I — Intelligence

S — System

These layers have deliberately different responsibilities.



MAX is the conversational personality and primary human interface, whereas A.E.G.I.S. is the underlying platform.

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## 2. Vision and Mission

### 2.1 Vision

Build a personal artificial intelligence that can:

1. Understand natural human communication.
2. Remember long-term personal context.
3. Perceive the surrounding digital and physical environment.
4. Reason about that environment.
5. Plan multi-step tasks.
6. Use software tools.
7. Control connected devices.
8. Act autonomously within predefined permissions.
9. Ask the user for approval when actions exceed those permissions.
10. Continuously expand through new device and software integrations.

The ultimate vision is not:

"A smarter chatbot."

It is:

**A personal intelligence layer between the user and the digital/physical world.**

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## 3. Core Design Philosophy

A.E.G.I.S. should follow six principles.

## 3.1 One intelligence, many bodies

The intelligence should not be tied to a particular computer.

Today:

MAX → PC

Tomorrow:

MAX → PC

MAX → Mac

MAX → iPhone

MAX → Vehicle

MAX → Robot

MAX → Wearable

The intelligence remains conceptually the same while its available capabilities change according to the connected device.

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## 3.2 The LLM is not the whole AI

A critical architectural rule is:

MAX ≠ LLM

MAX should be composed of:

MAX

|

├ Personality

├ Conversation

├ Reasoning

├ Memory

├ World Model

├ Perception

├ Agents

├ Tools

├ Permissions

├ Device Context

└ Automation

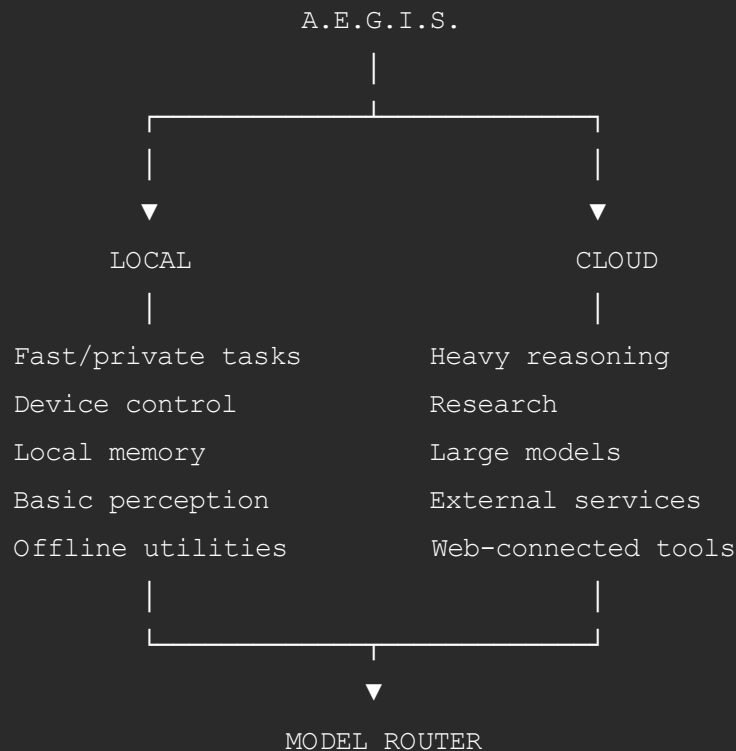
A language model is only one component within the Intelligence layer.

This makes it possible to replace one underlying model without rewriting the entire platform.

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### 3.3 Local-first, hybrid architecture

The personal version should prioritize privacy and local capabilities while still using internet and cloud services when they provide superior performance.



This model also creates a clean future path toward a paid cloud version without requiring the personal architecture to be discarded.

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## 4. User Requirements

The current project requirements are:

### 4.1 Personalization

A.E.G.I.S. is initially intended exclusively for the user.

The system should understand:

- preferred behavior
- routines

- projects
  - devices
  - communication style
  - frequently used applications
  - ongoing goals
  - historical context
  - personal knowledge
- 

## 4.2 Autonomy

The system is intended to progress through all five autonomy levels.

LEVEL 1

Reactive Assistant



LEVEL 2

Tool-Using Assistant



LEVEL 3

Proactive Assistant



LEVEL 4

Autonomous Agent



LEVEL 5

Persistent Autonomous Intelligence

The ultimate objective is Level 5.

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## 4.3 Voice personality

MAX should have a recognizable voice and conversational identity inspired by the general idea of sophisticated fictional AI assistants.

The implementation should remain original rather than reproducing a copyrighted character or voice.

MAX should eventually support:

- wake word activation
- natural conversational speech

- interruption
  - speaking while tasks execute
  - context-aware replies
  - multiple conversational modes
  - concise operational communication
  - detailed explanations when required
- 

## 4.4 Persistent memory

The user has explicitly chosen comprehensive personal memory.

Therefore memory is a first-class architectural component rather than a simple chat-history database.

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## 4.5 Maximum connected-world awareness

A.E.G.I.S. should eventually fuse as many authorized sensors and information sources as practically available:

Camera  
Microphone  
GPS  
IMU  
Computer state  
Phone sensors  
Network information  
Calendar  
Files  
Applications  
External sensors  
Vehicle telemetry  
Robotics sensors  
Environmental sensors

The practical objective is therefore:

**Maximum available situational awareness from authorized connected sources.**

Literal 100% awareness is impossible because no system can observe information it has no sensor or data source for. The architecture should nevertheless be designed to accept many heterogeneous sources.

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## 5. A.E.G.I.S. Five-Layer Architecture

### 5.1 A — Autonomous Layer

The Autonomous layer is responsible for converting objectives into actions.

A — AUTONOMOUS

- |
- |— Goal Manager
- |— Task Manager
- |— Planner
- |— Agent Runtime
- |— Workflow Engine
- |— Decision Engine
- |— Scheduler
- |— Retry / Recovery
- |— Self-Correction
- |— Multi-Step Execution
- |— Long-Horizon Planning

Its responsibility can be summarized as:

**"What should A.E.G.I.S. do next?"**

Example:

User:

**"Prepare everything for my CARLA development session."**

The Autonomous layer could generate:

1. Check computer state
2. Check required software
3. Start development services
4. Open IDE



5. Open project
6. Start CARLA
7. Validate Python environment
8. Check project status
9. Prepare relevant files
10. Report readiness

The planner should not directly perform every action. It proposes actions to the Guardian for authorization.

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## 5.2 E — Electronic Layer

The Electronic layer represents the physical and digital interfaces through which A.E.G.I.S. interacts with the world.

```
E — ELECTRONIC
|
├— PC
├— Mac
├— iPhone
├— Cameras
├— Microphones
├— GPS
├— IMU
├— ESP32
├— Raspberry Pi
├— Arduino
├— IoT
├— Vehicles
├— Robots
├— Wearables
└— Future Devices
```

Each device should expose standardized capabilities.

For example:

```
Computer
├— open_application()
├— close_application()
├— read_screen()
├— type_text()
└— move_mouse()
```

└─ execute\_command()

Phone

└─ get\_location()

└─ capture\_image()

└─ access\_sensor()

└─ send\_notification()

Vehicle

└─ read\_vehicle\_state()

└─ get\_sensor\_data()

└─ navigation()

└─ authorized\_control()

The AI should not care about the implementation details of each device.

That responsibility belongs to device adapters.

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## 5.3 G — Guardian Layer

The Guardian layer is one of the most important components.

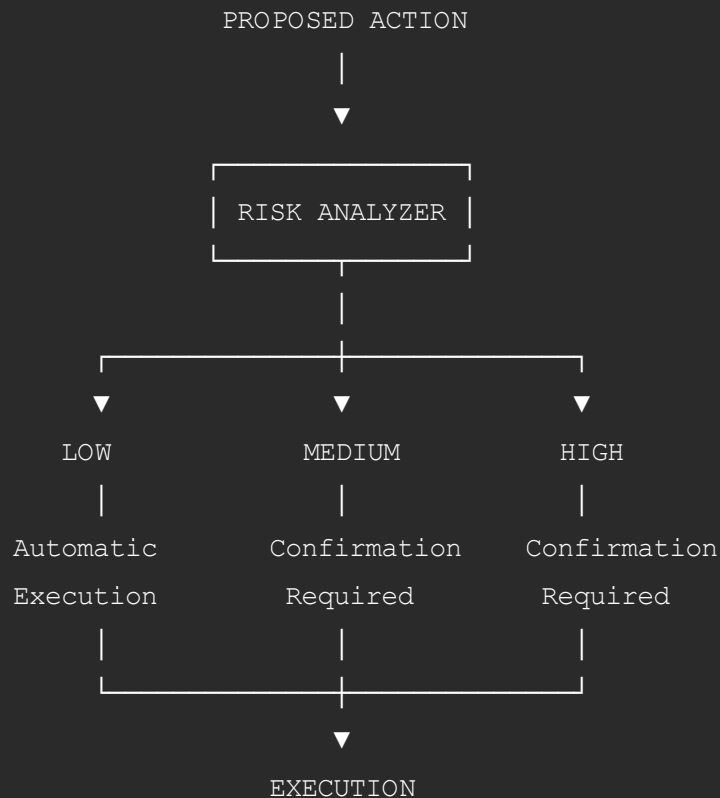
It is responsible for:

- safety
- authorization
- authentication
- risk analysis
- user confirmation
- device restrictions
- rate limits
- emergency stop
- auditing
- anomaly detection
- policy enforcement

The core principle is:

**The AI proposes actions; the Guardian decides whether those actions may execute.**

## Risk architecture



Examples:

### Low risk

- open applications
- search the web
- read files
- create notes
- change volume
- organize low-risk files
- launch known development environments

### Medium risk

- send a message
- modify important files
- install software
- publish information
- create appointments
- change meaningful system configuration

### High risk

- financial transactions
- deleting critical data
- security-system actions
- physical machinery
- vehicle control
- aircraft control
- dangerous physical actions

For high-risk systems, additional subsystem-specific safety controllers should eventually exist below Guardian.

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## 5.4 I — Intelligence Layer

The Intelligence layer is the cognitive center.

```
I — INTELLIGENCE
|
├— Language Model
├— Reasoning
├— Speech Recognition
├— Speech Generation
├— Computer Vision
├— Multimodal Understanding
├— Memory
├— Context Engine
├— Knowledge Retrieval
├— World Model
├— User Model
├— Sensor Fusion
└— Model Router
```

The system should not always use the same model.

A model router may eventually choose:

```
Simple request
    ↓
Fast model

Complex reasoning
    ↓
Large reasoning model
```

Private task

↓

Local model

Vision task

↓

Vision model

Speech

↓

Speech model

This improves efficiency and makes the architecture vendor-independent.

Apple's current Foundation Models framework is one example of the industry moving toward model abstraction, on-device models, multimodal inputs and tool calling, including the ability to combine model configurations and tools for agentic applications.

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## 5.5 S — System Layer

The System layer connects everything.

S — SYSTEM

|

├ API Gateway

├ Event Bus

├ Device Registry

├ Service Discovery

├ Networking

├ Message Routing

├ Authentication

├ Configuration

├ Logging

├ Metrics

├ Distributed State

├ Version Management

└ Deployment

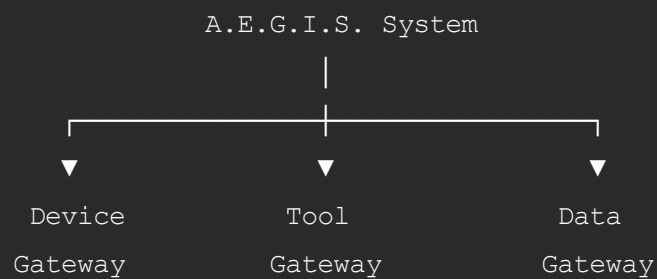
This layer prevents every subsystem from directly talking to every other subsystem.

Without it:

AI → PC  
AI → phone  
AI → database  
AI → camera  
AI → sensor  
AI → web  
AI → robot  
AI → car  
...

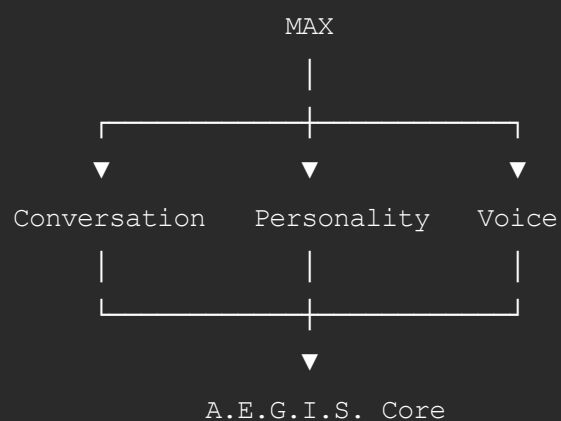
The architecture rapidly becomes spaghetti with Ethernet cables.

With the System layer:



## 6. MAX Architecture

MAX should be the personality and interaction layer.



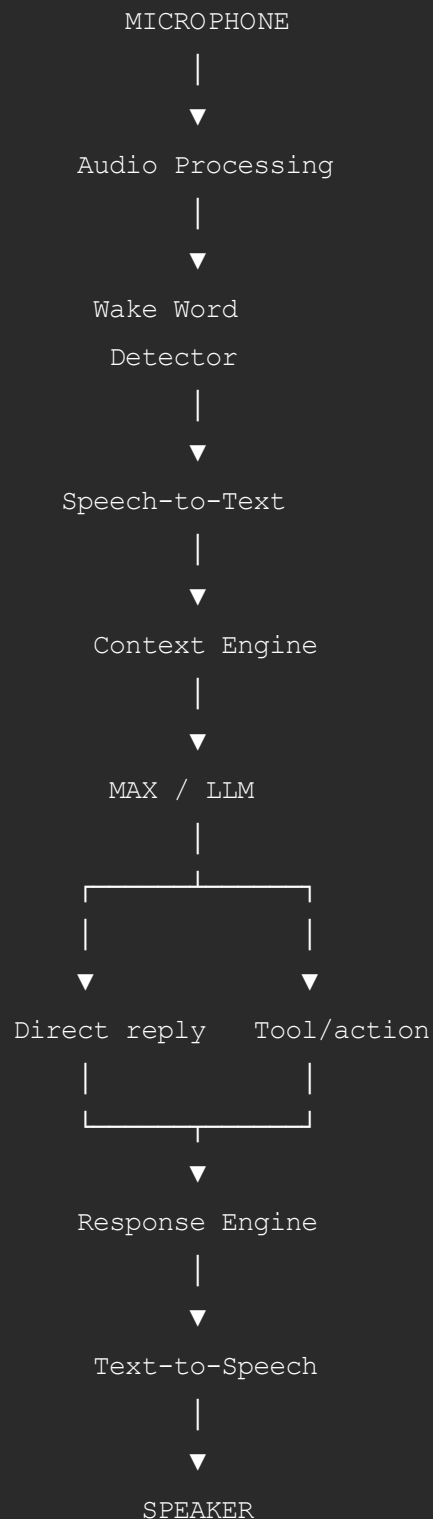
MAX should eventually understand:

- who is speaking
- what was discussed
- what is happening
- what the user is trying to accomplish

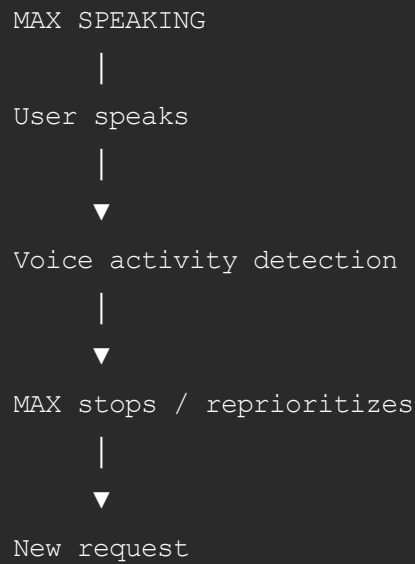
- what tools are available
  - what actions are permitted
  - which previous information is relevant
- 

## 7. Voice Pipeline

The initial voice architecture should be:



Eventually the conversation loop should support interruption:

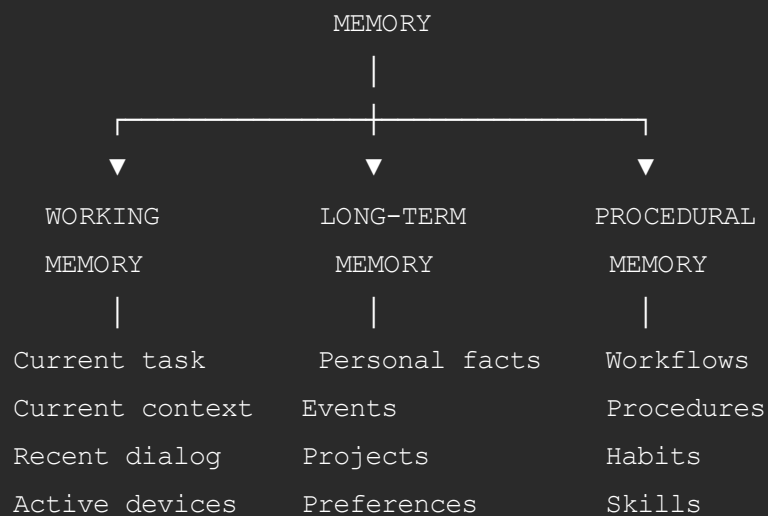


This is necessary for the system to feel conversational rather than like a walkie-talkie with a PhD.

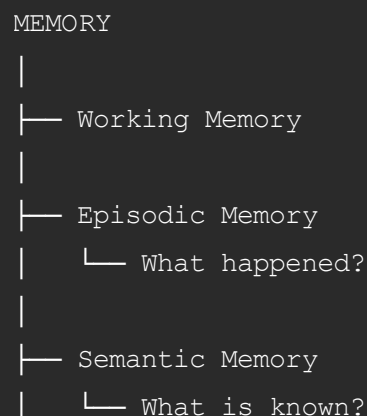
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## 8. Persistent Memory Architecture

A.E.G.I.S. should use multiple forms of memory.



A more complete structure:





- |
- |— Procedural Memory
  - | — How is something done?
- |
- |— Preference Memory
  - | — What does the user prefer?
- |
- |— Project Memory
  - | — What is being built?
- |
- |— Relationship/Entity Memory
  - | — People, devices, places, objects

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## 9. Memory Governance

Complete memory requires equally serious memory controls.

The user should eventually be able to:

- View memory
- Search memory
- Edit memory
- Delete memory
- Forget specific information
- Export memory
- Disable categories of memory
- View memory source
- View when memory was created

Every remembered item should ideally have provenance.

For example:

Memory:  
"User prefers Python."

Source:  
Conversation

Created:  
2026-08-19

Confidence:

High

Last confirmed:

2026-08-19

This reduces the risk of treating an old assumption as permanent truth.

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## 10. Context Engine

Memory alone is insufficient.

MAX needs to distinguish between:

Something that is remembered

and

Something that is relevant right now.

The Context Engine should merge:

Current conversation

+

Current task

+

Current device

+

Current location

+

Current time

+

Recent history

+

Relevant long-term memories

+

Active projects

+

Available tools

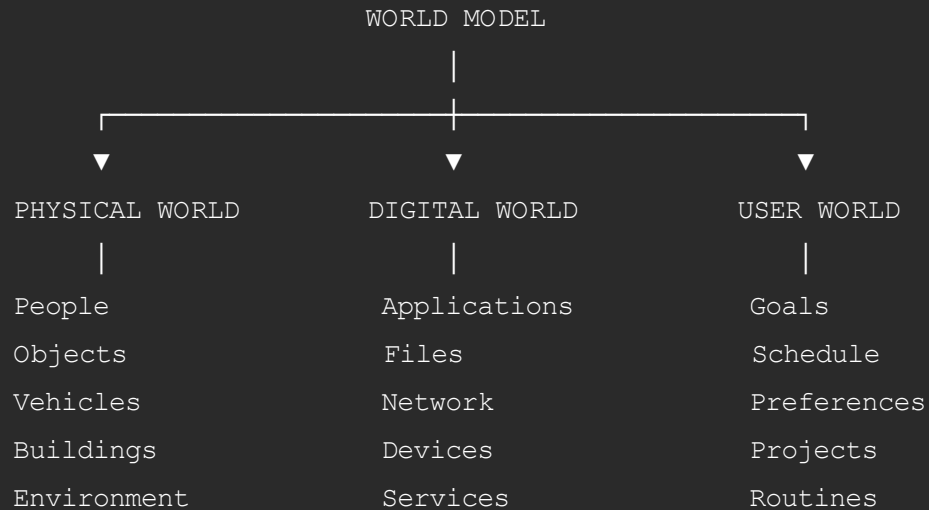
+

Environmental information

into a temporary context representation.

## 11. World Model

The World Model is one of the defining features of the long-term A.E.G.I.S. system.



The World Model should maintain entities, relationships, states, locations and events.

Example:

Person

- └ identity
- └ location
- └ activity
- └ relationship

Device

- └ owner
- └ location
- └ state
- └ capabilities
- └ permissions

Vehicle

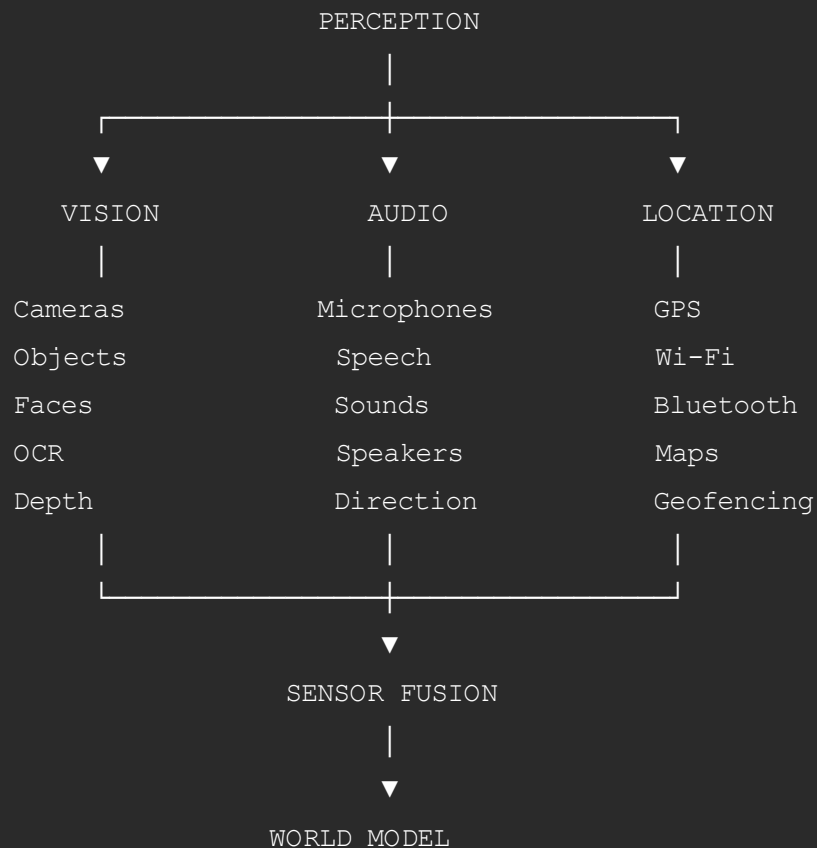
- └ location
- └ speed
- └ sensor state
- └ navigation state
- └ autonomy state

This representation is what eventually allows higher-level reasoning.

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## 12. Perception Architecture

The perception layer will eventually process multiple modalities.



## 13. Sensor Fusion

Sensor fusion should become a fundamental architecture rather than a later add-on.

The purpose is to combine multiple imperfect observations into a more reliable state estimate.

For example:

Camera says:

"Object approximately 5 m away"

IMU says:

"Device moving"

GPS says:

"Location = office entrance"

Microphone says:

"Human speech detected"

Previous state says:

"Known person expected nearby"



SENSOR FUSION



"Known person approaching user  
at office entrance while user is moving."

**For future robotics, the architecture can become:**

Camera

LIDAR

Radar

GPS

IMU

Wheel odometry

Map

Vehicle telemetry



Sensor Fusion



Perception State



World Model



Planning / Control

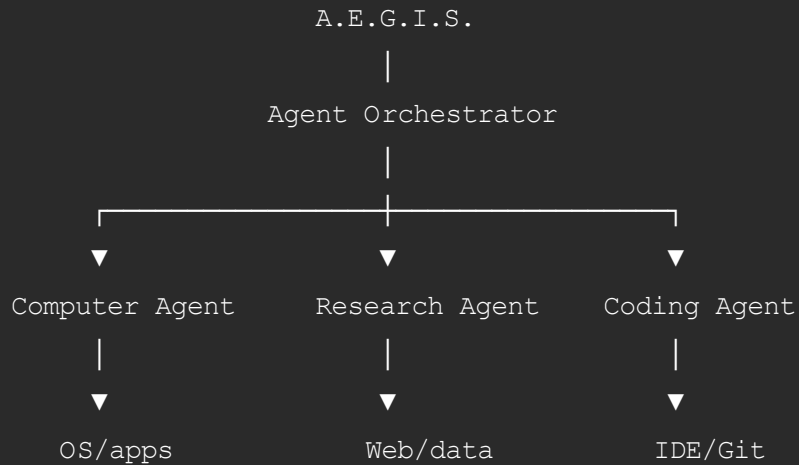
CARLA is particularly useful for this development because its sensor system can stream simulated sensor data from cameras,

detectors and other sensor actors into driving agents.

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## 14. Agent Architecture

A.E.G.I.S. should use specialist agents instead of one enormous agent.



Additional future agents:

Vehicle Agent  
Navigation Agent  
Communication Agent  
Personal Assistant Agent  
Vision Agent  
Robotics Agent  
Home Automation Agent  
Security Agent  
System Administration Agent

The Orchestrator decides which specialist should be used.

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## 15. Tool Architecture

Tools provide the bridge between intelligence and action.

Conceptually:



- |
- └─ Browser
- └─ Files
- └─ Terminal
- └─ Applications
- └─ Calendar
- └─ Database
- └─ Communication
- └─ Vision
- └─ Sensors
- └─ Devices

Tool metadata should include:

- Name
- Description
- Parameters
- Permissions
- Risk category
- Required device
- Required authentication
- Expected result
- Timeout
- Rollback strategy

This becomes important because an AI should not merely know **what a tool does**. It should know **when it is allowed to use it**.

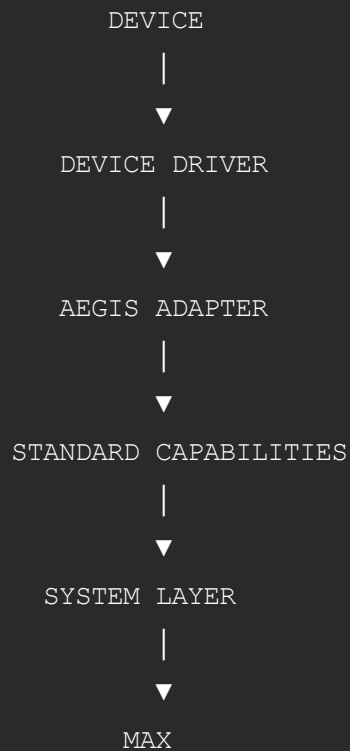
MCP is especially relevant here because its current specification standardizes ways for AI applications to connect with tools, resources and prompts, while its July 2026 specification added further authorization hardening, routing and stateless protocol improvements.

A.E.G.I.S. should be MCP-compatible rather than reinventing every tool integration.

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## 16. Device Abstraction

Every connected device should expose capabilities through a standardized adapter.



### Example:

Windows PC



Windows Adapter



open\_app()  
read\_screen()  
execute\_command()

Mac



macOS Adapter



open\_app()  
read\_screen()  
execute\_command()

Future vehicle



Vehicle Adapter



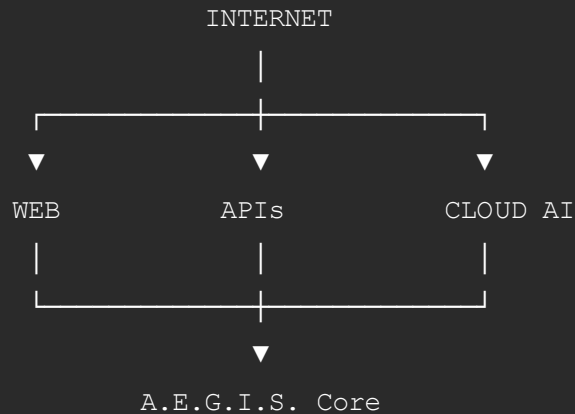
read\_speed()  
read\_environment()  
set\_authorized\_mode()

This is what allows "one AI, many bodies."



## 17. Internet Architecture

A.E.G.I.S. should assume internet availability for full functionality.



Without internet, limited local operations can continue:

Offline

- |
- |— Open application
- |— Read local state
- |— Change safe local settings
- |— Execute local workflows
- |— Access locally cached memory

Internet-dependent capabilities should degrade gracefully rather than crash the entire system.

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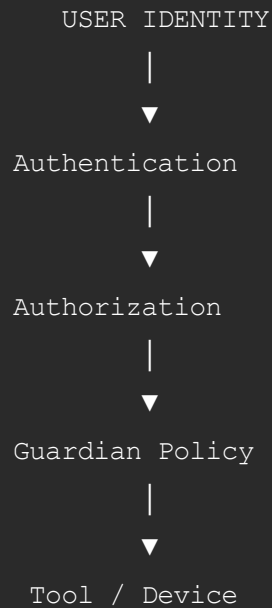
## 18. Security Architecture

Security should not be a later feature.

The first version should already separate:

- Identity
- Authentication
- Authorization
- Secrets
- Tools
- Memory
- Devices
- Logs

The future architecture should include:



Secrets should never be placed directly in prompts.

API keys, passwords, tokens and credentials should live in secure secret storage and be accessed by controlled connectors.

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## 19. Audit Architecture

Every meaningful autonomous action should generate an audit event.

Example:

Timestamp:  
2026-08-19 01:23:12

Agent:  
Computer Agent

Requested Action:  
Open Visual Studio Code

Risk:  
LOW

Guardian Decision:  
ALLOW

Execution:

SUCCESS

For a medium-risk action:

Requested Action:

Send message

Risk:

MEDIUM

Guardian:

CONFIRMATION\_REQUIRED

User:

APPROVED

Execution:

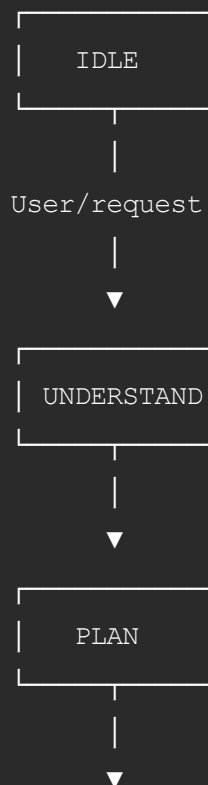
SUCCESS

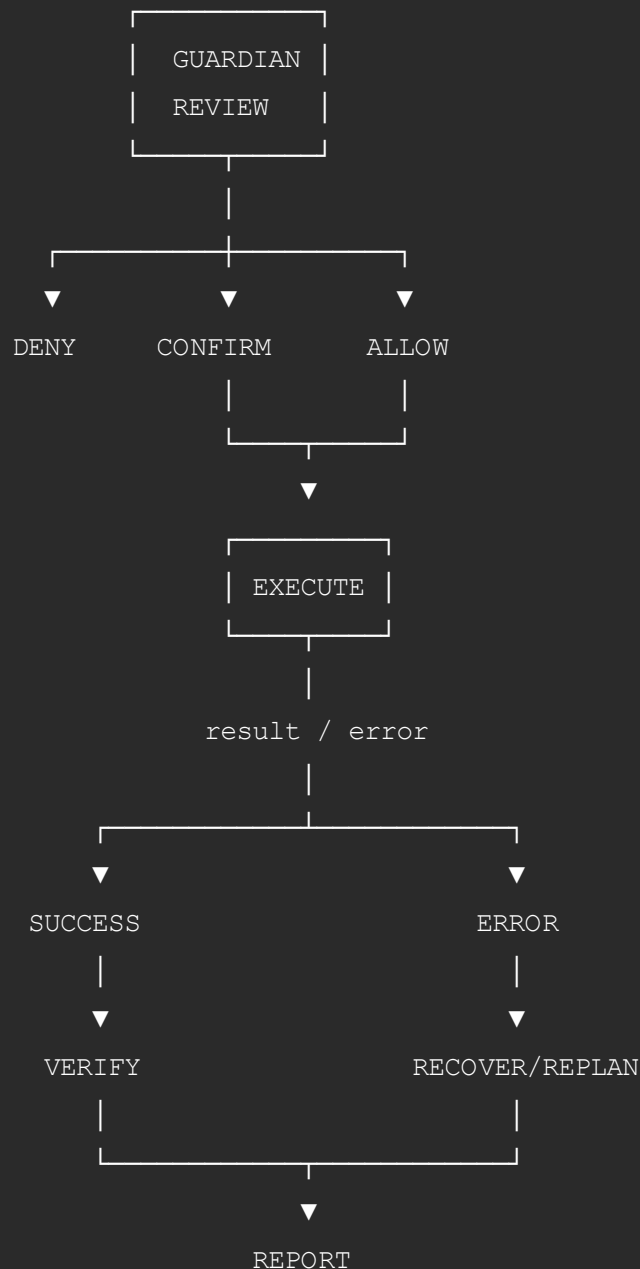
This becomes critical when autonomy expands.

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## 20. Autonomy State Machine

A.E.G.I.S. can model autonomy through a state machine.



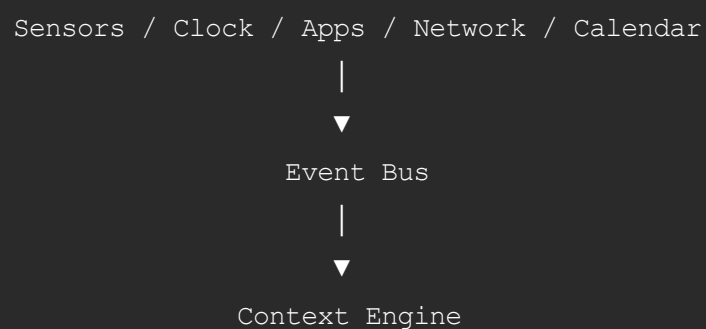


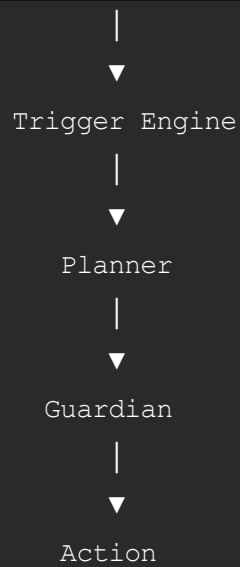
## 21. Proactive Behavior

Level 3 and above require event-driven behavior.

A.E.G.I.S. should not only wait for user prompts.

An event architecture could look like:





### Example:

Event:

Laptop battery low

Context:

Important process running

Analysis:

Charging required

Action:

Notify user

### Later:

Event:

Scheduled development session begins

Context:

CARLA project task pending

Action:

Prepare environment

---

## 22. Self-Correction

Level 4 and 5 require recovery.

Suppose:

MAX:

Start CARLA

Execution:

FAIL

System:

Error detected

Planner:

Determine cause

Possible action:

Restart service

Guardian:

LOW RISK → ALLOW

Execution:

Retry

Result:

SUCCESS

**This creates a feedback loop:**

PLAN

↓

EXECUTE

↓

OBSERVE

↓

EVALUATE

↓

CORRECT

↓

EXECUTE AGAIN

The system should have bounded retries and failure escalation so it does not spend three hours restarting the same broken service because a language model felt optimistic.

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## 23. Computer Control

The PC should be the first "body" of A.E.G.I.S.

Initial capabilities:

Application control

Window management

Keyboard

Mouse

Terminal

Files

Browser

System information

Screen capture

Process management

Notifications

The computer agent should initially work in a controlled environment.

The long-term vision is:

User

↓

MAX

↓

Computer Agent

↓

Guardian

↓

Windows/macOS Adapter

↓

OS

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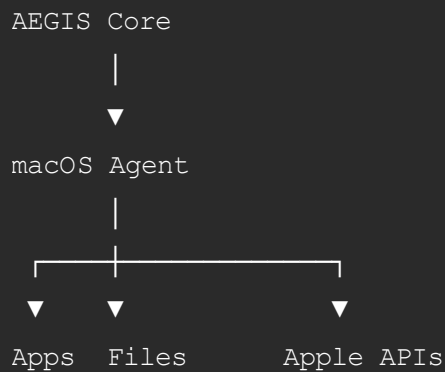
## 24. Mac Integration

The Mac should become a second major development and execution environment.

The Apple ecosystem is increasingly useful for this architecture. Apple's current Foundation Models framework supports on-device language models, multimodal prompts, tool calling and dynamic

model/tool configurations, with additional support for cloud and other model providers.

Therefore the eventual architecture can include:



The M5 Mac can also become an important edge-AI test platform.

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## 25. iPhone Integration

The iPhone should eventually become an **edge perception and interaction node**, not just a remote control.

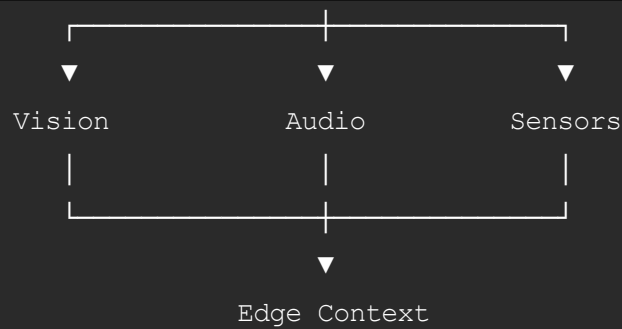
Potential capabilities:

Camera  
Microphones  
GPS  
Motion sensors  
Notifications  
Context  
Voice  
Mobile interface  
Network

Conceptually:







Apple currently provides on-device model and tool-calling APIs that make this kind of hybrid architecture increasingly practical.

## 26. Android Future Compatibility

Although the current personal device is an iPhone, the commercial platform should remain device-agnostic.

Android's current AppFunctions direction is particularly relevant because it allows applications to expose capabilities to AI agents as system-discoverable functions, with permission requirements around execution.

This supports the future vision:



## 27. Robotics and Future Physical Systems

Robotics should initially remain a later phase.

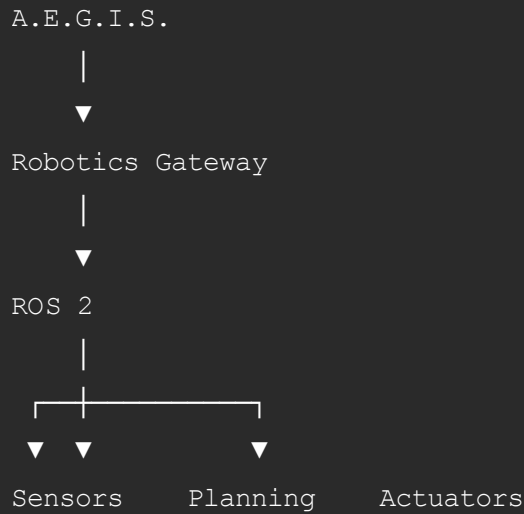
The software architecture should nevertheless prepare for it now.

ROS 2 provides communication primitives particularly relevant to future A.E.G.I.S. robotics integration:

- **Topics** for continuous streams such as sensor data.
- **Services** for short request/response interactions.

- **Actions** for long-running operations with feedback and cancellation.

That maps naturally onto A.E.G.I.S.:



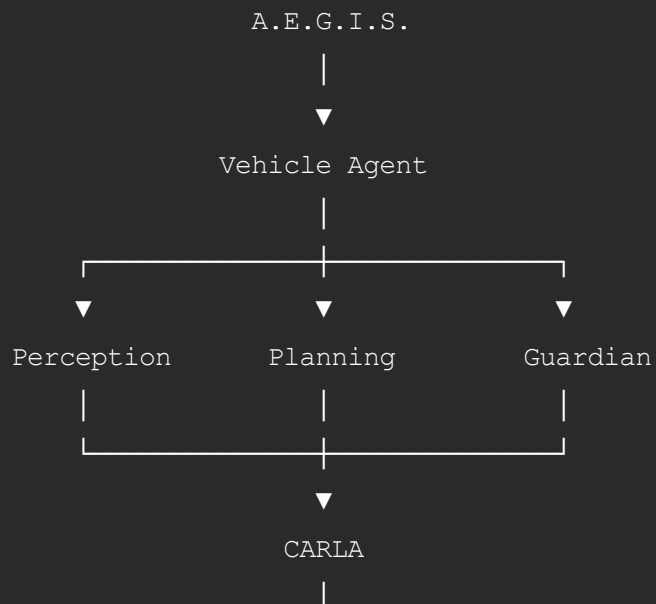
ROS 2 nodes can operate in the same process, different processes, or different machines, which suits the eventual distributed architecture.

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## 28. CARLA Integration

CARLA should become a major intermediate milestone because it allows testing automotive intelligence without placing an actual vehicle or innocent bystander in the feedback loop.

The architecture:



CARLA exposes simulated sensors as actors that can stream environmental data during simulation, making it suitable for perception and sensor-fusion development.

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## 29. Current Hardware Allocation

### RTX 3060 PC

Primary:

AI laboratory  
Local model testing  
Vision  
Speech experiments  
Development  
CARLA  
Simulation  
GPU workloads

Its 12 GB VRAM makes it useful for local inference and experimentation, although it should not be treated as unlimited compute.

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### M5 MacBook

Primary:

Software development  
macOS integration  
Apple ecosystem  
Mobile development  
Edge-AI experiments  
Secondary model workloads

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### Raspberry Pi 3

Future:

Edge gateway  
Sensor hub  
IoT  
Robotics experiments  
Physical-world interface

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## ESP32

### Future:

Distributed sensors  
Wi-Fi/Bluetooth devices  
Environmental telemetry  
Actuators  
Embedded control

---

## Arduino Uno

### Future:

Basic electronics  
Simple sensor prototypes  
Actuator control  
Educational hardware experiments

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## 30. Proposed Software Stack

The first architecture should intentionally minimize language complexity.

### Primary

#### Python

#### Used for:

- A.E.G.I.S. services
- AI orchestration
- APIs
- memory

- tool system
- computer vision
- experimentation
- automation

## Secondary

### C/C++

Used when required for:

- embedded systems
- performance-critical components
- robotics
- hardware integration

## Later

### TypeScript

Used for:

- web dashboard
- advanced UI
- cross-platform interfaces

### Swift

Used for:

- deep iOS/macOS integration

### Kotlin

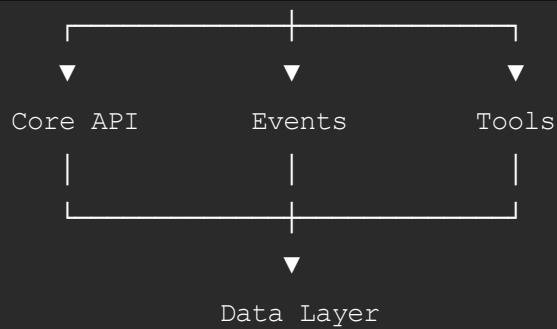
Used if an Android client is eventually developed.

---

## 31. Backend Architecture

Initial backend:





Likely initial technologies:

Python  
FastAPI  
WebSocket  
PostgreSQL  
Redis  
Vector storage  
Docker  
Git

These should not all be introduced on day one. Complexity must grow with the system rather than arrive as a ceremonial offering to the software gods.

---

## 32. Event-Driven Architecture

A.E.G.I.S. should increasingly become event driven.

Example:

EVENT:

User enters office

↓

Context Engine

↓

Relevant state:

- User detected
- PC available
- Scheduled task exists

```
↓  
  
Planner  
  
↓  
  
Action proposal  
  
↓  
  
Guardian  
  
↓  
  
Execute
```

This is more scalable than having every component constantly polling every other component.

---

## 33. Repository Architecture

A proposed repository structure:

```
aegis/  
|  
├─ README.md  
├─ pyproject.toml  
├─ .env.example  
├─ docker-compose.yml  
|  
├─ apps/  
|   ├─ max/  
|   ├─ desktop/  
|   └─ api/  
|  
├─ core/  
|   ├─ intelligence/  
|   ├─ autonomy/  
|   ├─ guardian/  
|   ├─ system/  
|   └─ electronic/  
|
```

```
|— agents/
|   |— computer/
|   |— research/
|   |— coding/
|   |— communication/
|   └— future/
|
|— perception/
|   |— vision/
|   |— speech/
|   |— audio/
|   |— location/
|   └— fusion/
|
|— memory/
|   |— working/
|   |— episodic/
|   |— semantic/
|   |— procedural/
|   └— retrieval/
|
|— tools/
|   |— browser/
|   |— filesystem/
|   |— terminal/
|   |— applications/
|   └— devices/
|
|— devices/
|   |— windows/
|   |— macos/
|   |— ios/
|   |— raspberrypi/
|   |— esp32/
|   └— future/
|
|— security/
|   |— auth/
|   |— permissions/
|   |— secrets/
|   └— audit/
|
|— world/
```



```
|   ├── entities/
|   ├── state/
|   ├── context/
|   └── events/
|
|── tests/
|
|── docs/
|   ├── architecture/
|   ├── api/
|   ├── security/
|   └── development/
|
|── scripts/
```

---

## 34. Development Strategy

Do not build the entire system simultaneously.

The correct progression is:

```
STAGE 0
Architecture
    ↓
STAGE 1
Conversation
    ↓
STAGE 2
Voice
    ↓
STAGE 3
Tools
    ↓
STAGE 4
Computer control
    ↓
STAGE 5
Memory
    ↓
STAGE 6
Agents
    ↓
```

STAGE 7  
Autonomy  
↓  
STAGE 8  
Perception  
↓  
STAGE 9  
World Model  
↓  
STAGE 10  
Sensor Fusion  
↓  
STAGE 11  
Multi-device  
↓  
STAGE 12  
CARLA  
↓  
STAGE 13  
Physical systems

---

## 35. A.E.G.I.S. Version Roadmap

### v0.1 — Core Brain

#### Capabilities:

Text conversation  
LLM integration  
Basic personality  
Command handling  
Basic architecture

---

### v0.2 — Voice

Wake word  
Speech-to-text  
Text-to-speech  
Conversation loop

---

## **v0.3 — Tools**

Browser  
Files  
Terminal  
Applications  
System information

---

## **v0.4 — Computer Agent**

Open applications  
Read screen  
Keyboard/mouse  
Process management  
File workflows

---

## **v0.5 — Persistent Memory**

Working memory  
Long-term memory  
Semantic retrieval  
Personal preferences  
Project memory  
Memory management

---

## **v0.6 — Agent System**

Computer Agent  
Research Agent  
Coding Agent  
Communication Agent  
Agent orchestrator

---

## **v0.7 — Guardian**

Risk classifier  
Confirmation system  
Permission management

Audit logs  
Secrets management

The Guardian should exist before serious autonomy.

---

## **v0.8 — Proactive Intelligence**

Events  
Schedules  
Triggers  
Notifications  
Background context  
Suggested actions

---

## **v0.9 — Vision**

Camera input  
Image understanding  
OCR  
Object recognition  
Scene understanding

---

## **v1.0 — Personal A.E.G.I.S.**

Voice  
Memory  
Reasoning  
Tools  
Agents  
Autonomy  
Guardian  
Vision  
Context  
Computer control

This is the first major milestone.

---

## **36. v1.5 — World Model**

People  
Places  
Devices  
Projects  
Applications  
Objects  
Events  
Relationships  
Current state

---

## 37. v2.0 — Sensor Fusion

Vision  
Audio  
GPS  
Device state  
Sensors  
Context  
Environment

Unified into a world model.

---

## 38. v3.0 — Multi-Device A.E.G.I.S.

PC  
Mac  
iPhone  
Raspberry Pi  
ESP32  
Other devices

---

## 39. v4.0 — CARLA Autonomy

Perception  
Sensor fusion  
Planning  
Vehicle agent  
Simulation  
Autonomous driving experiments

## 40. v5.0 — Physical A.E.G.I.S.

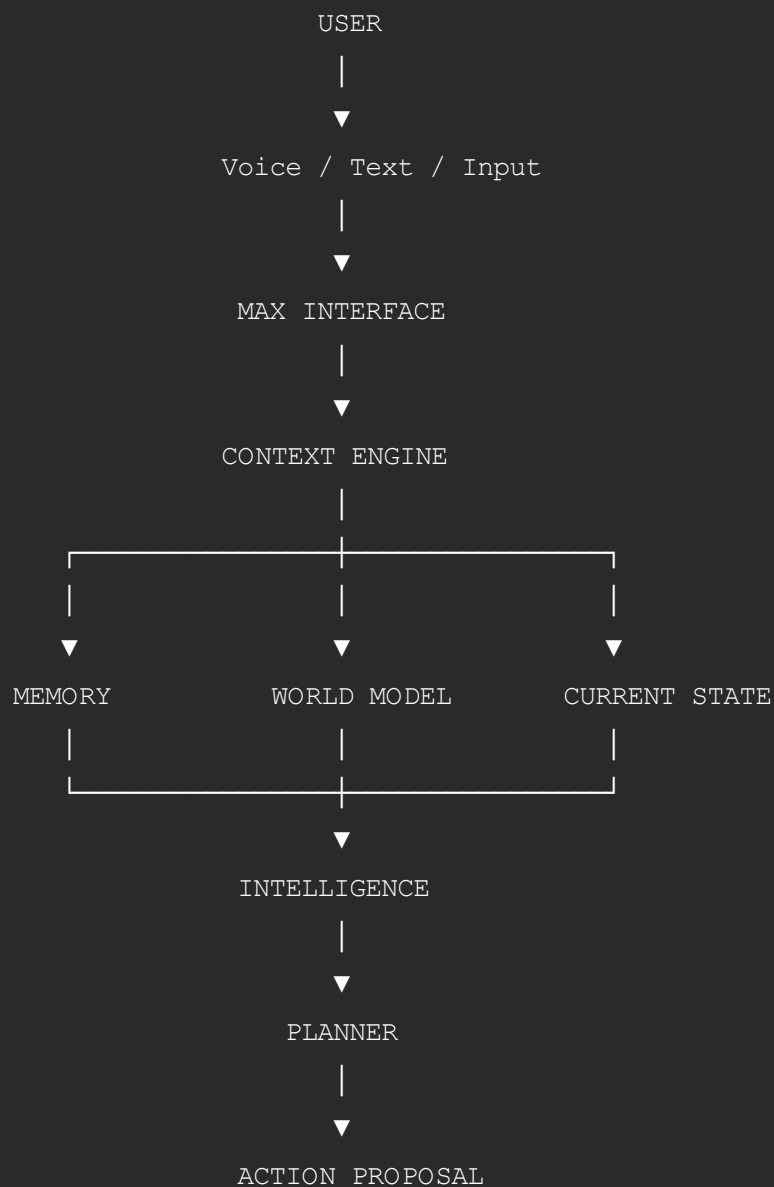
Robotics  
Vehicles  
Embedded systems  
Wearables  
Real-world sensors  
Physical autonomy

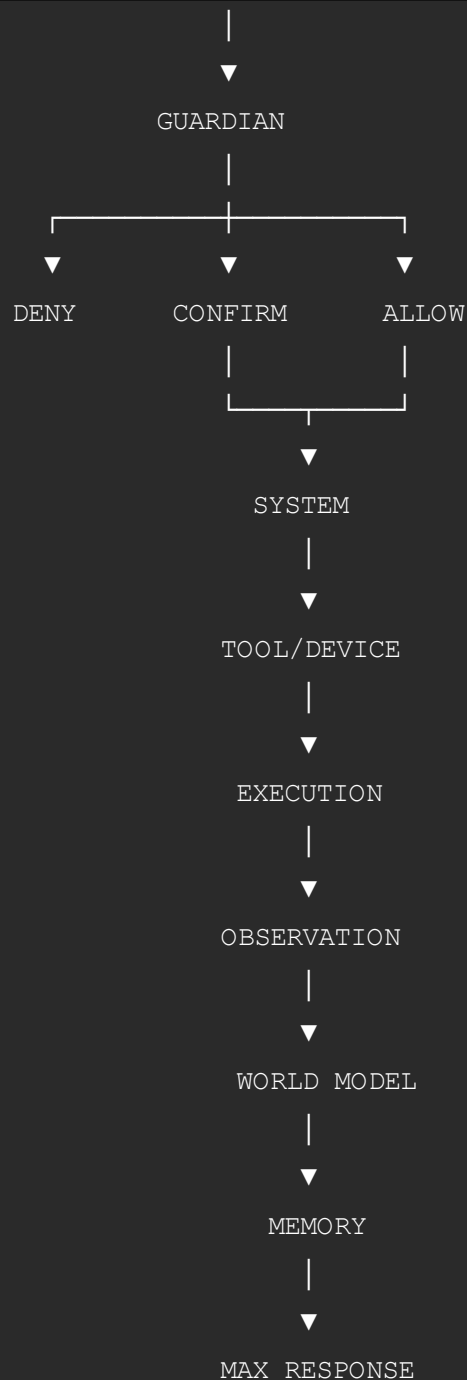
This should be treated as a separate engineering domain with additional safety requirements.

---

## 41. High-Level End-to-End Flow

The fundamental A.E.G.I.S. request pipeline:

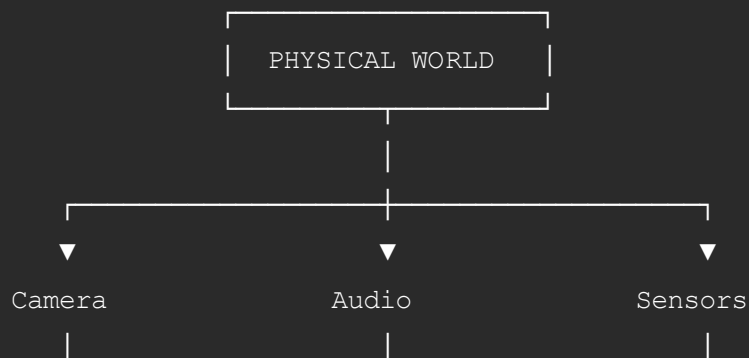


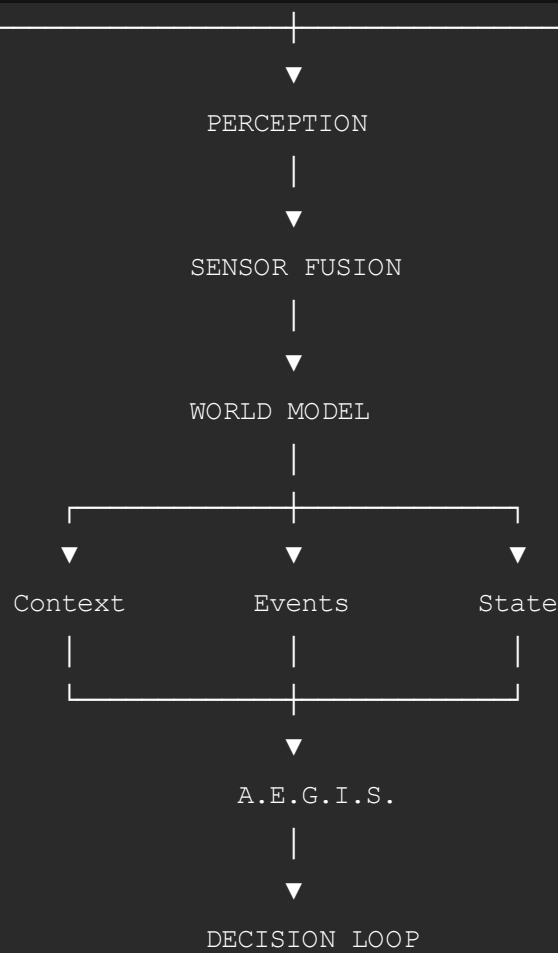


---

## 42. Continuous World-Awareness Flow

Unlike a chatbot, A.E.G.I.S. should eventually operate continuously.



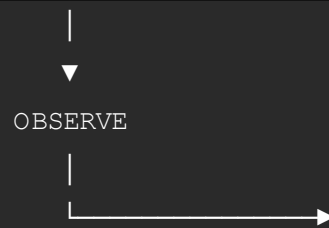


---

## 43. Autonomous Decision Loop







This closed loop is the foundation of Level 5 autonomy.

---

## 44. Example: Simple User Interaction

User:

"MAX, open my CARLA project."

Pipeline:

```
Voice
  ↓
Speech-to-text
  ↓
MAX
  ↓
Context
  ↓
Project memory
  ↓
Identify CARLA project
  ↓
Computer Agent
  ↓
Guardian
  ↓
LOW RISK
  ↓
Execute
  ↓
Open IDE/project
  ↓
Verify
  ↓
Respond
```

---

## 45. Example: Complex Autonomous Workflow

User:

"Prepare everything for tomorrow's autonomous-driving work."

MAX might:

1. Identify relevant project.
2. Check project status.
3. Inspect required software.
4. Check CARLA installation.
5. Check Python environment.
6. Check GPU availability.
7. Inspect recent notes.
8. Identify unfinished tasks.
9. Prepare the development environment.
10. Generate a short task summary.

The user doesn't manually specify every operation.

This is where A.E.G.I.S. begins approaching Level 4 behavior.

---

## 46. Example: Medium-Risk Action

User:

"Message the team that the simulation is ready."

Pipeline:

```
Understand request
    ↓
Find communication tool
    ↓
Prepare message
    ↓
Guardian
    ↓
MEDIUM RISK
    ↓
```

Ask confirmation



User approves



Send



Record action

---

## 47. Example: High-Risk Physical Action

Future:

"Drive me home."

A language model should not directly control vehicle actuators.

Instead:

MAX



Vehicle Agent



Guardian



Vehicle Safety Controller



Perception / Planning



Validated Command



Vehicle Control

This introduces a crucial principle:

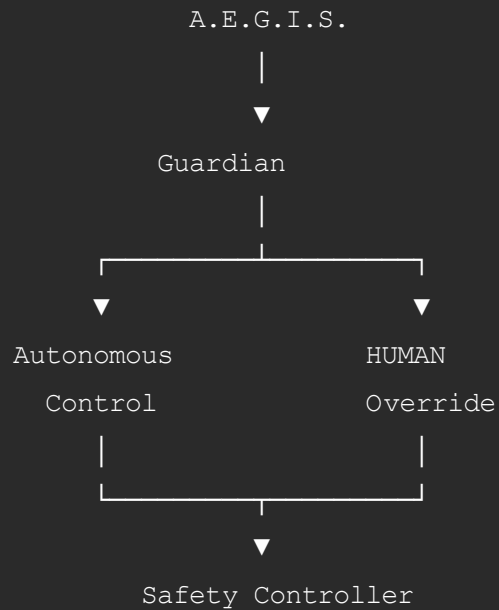
**General-purpose AI can request high-level actions, but safety-critical systems must retain deterministic and independently validated control layers.**

This distinction becomes increasingly important when the project expands from software automation into physical autonomy.

---

## 48. Human Override

Every physical system should eventually support an independent human override.



Emergency stop mechanisms should not depend solely on the same AI process they are intended to interrupt.

---

## 49. Observability

A large autonomous system needs visibility into itself.

A.E.G.I.S. should eventually expose:

- CPU
- GPU
- Memory
- Network
- Latency
- Model usage
- Tool usage
- Agent state
- Errors
- Sensor health
- Device health
- Action history
- Guardian decisions

A developer dashboard could look like:

| A.E.G.I.S.                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------|
| MAX: ONLINE                                                                                                                 |
| Guardian: ACTIVE                                                                                                            |
| Memory: HEALTHY                                                                                                             |
| World Model: ACTIVE                                                                                                         |
| PC: ONLINE                                                                                                                  |
| Mac: ONLINE                                                                                                                 |
| Internet: CONNECTED                                                                                                         |
| Vision: READY                                                                                                               |
| Voice: READY                                                                                                                |
| ACTIVE TASK                                                                                                                 |
| Preparing CARLA environment                                                                                                 |
| RECENT EVENTS                                                                                                               |
| <ul style="list-style-type: none"> <li>• Project opened</li> <li>• GPU detected</li> <li>• CARLA server starting</li> </ul> |

## 50. Failure Handling

A.E.G.I.S. must be designed assuming things will fail.

Potential failures:

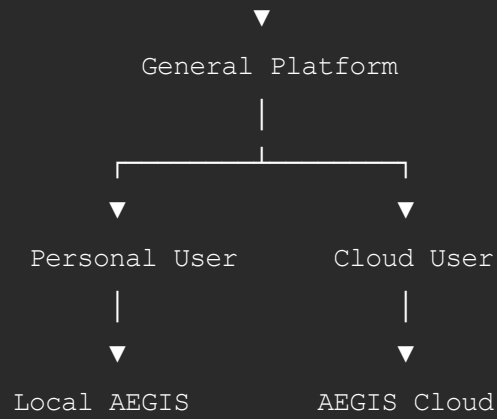
```

Internet unavailable
Model unavailable
GPU overloaded
Tool failure
Device disconnected
Sensor failure
Incorrect perception
Permission denied
User rejects action
Agent loop
Memory corruption
Unexpected application state

```

The architecture therefore needs:





The future commercial platform could contain:

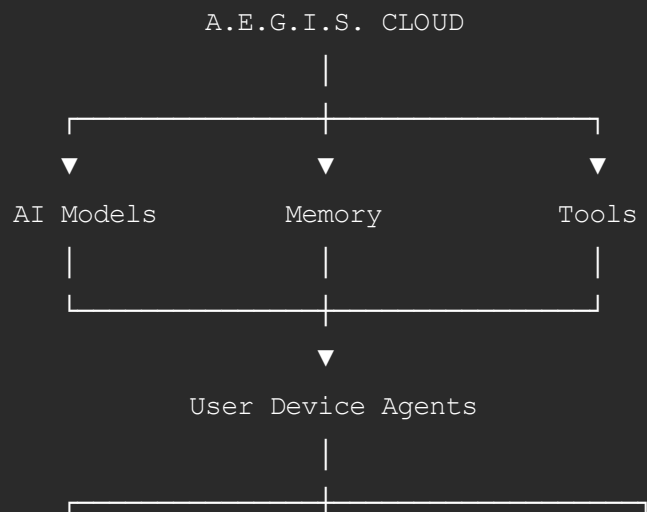
Identity  
User account  
Personal memory vault  
Cloud models  
Device registry  
Agent marketplace  
Tool integrations  
Billing  
Usage management  
Security  
Team/workspace features

The current architecture should therefore avoid hard-coding the project exclusively around one machine.

---

## 53. Future SaaS Model

Potential long-term structure:



▼  
PC

▼  
Phone

▼  
Car

This could eventually support different subscription levels, compute allocation, storage, device integrations and advanced autonomy.

---

## 54. Research Platform Potential

A.E.G.I.S. can also become a research platform.

Research areas could include:

Multimodal AI  
Agentic systems  
Long-term memory  
Personal AI  
Sensor fusion  
World models  
Human-AI interaction  
Edge AI  
Robotics  
Autonomous vehicles  
Distributed intelligence  
AI safety  
Adaptive planning

That makes the project more valuable academically than simply creating a personal chatbot.

---

## 55. Company Potential

The same foundation can become a product architecture.

Potential product directions:

Personal AI  
Developer AI  
Vehicle AI  
Robotics AI  
Industrial AI  
Edge AI



AI operating environment

Enterprise autonomous agent platform

The company opportunity would come later.

The engineering objective now should be to create a technically sound core rather than prematurely building billing pages for a product that does not yet know how to open Notepad.

---

## **56. Main Technical Risks**

### **Risk 1 — Scope explosion**

The vision is enormous.

Solution:

Core first

→ desktop

→ memory

→ agents

→ perception

→ multi-device

→ physical systems

---

### **Risk 2 — Overdependence on LLMs**

Solution:

Use specialized deterministic subsystems where appropriate.

---

### **Risk 3 — Uncontrolled autonomy**

Solution:

Guardian architecture from the beginning.

---

### **Risk 4 — Memory becoming inaccurate**

Solution:

Memory provenance, confidence and user controls.

---

## **Risk 5 — System becoming impossible to maintain**

Solution:

Strict modularity and service boundaries.

---

## **Risk 6 — Hardware complexity arriving too early**

Solution:

Software-first development.

---

## **Risk 7 — Security**

Solution:

Treat tools, credentials, devices and memory as separate security domains.

---

# **57. What Should NOT Be Built First**

The following should intentionally wait:

- ✗ Real autonomous car control
- ✗ Aircraft control
- ✗ Full robotics autonomy
- ✗ Massive distributed sensor network
- ✗ Custom neural network from scratch
- ✗ Proprietary foundation model
- ✗ Custom operating system kernel
- ✗ Commercial cloud platform
- ✗ Huge multi-agent swarm

These are future layers.

The first objective is to make A.E.G.I.S. **actually useful on one computer.**

---

## 58. What Should Be Built First

The initial product should be:

A.E.G.I.S. v0.1

Desktop Personal AI

Capabilities:

- ✓ Conversation
- ✓ MAX personality
- ✓ Voice
- ✓ Wake word
- ✓ Local + cloud model routing
- ✓ Computer control
- ✓ Browser
- ✓ Files
- ✓ Terminal
- ✓ Memory
- ✓ Basic tools
- ✓ Guardian
- ✓ Audit logs
- ✓ Configuration

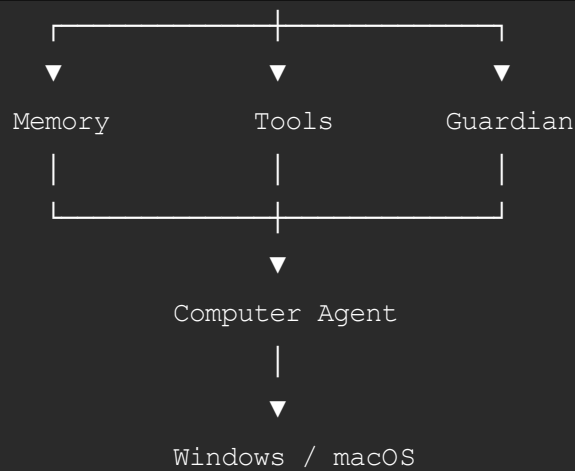
The system should be installable and startable as a normal application.

---

## 59. First Practical Development Architecture

The initial runtime could be:





No robotics.

No vehicle.

No gigantic distributed cluster.

Just a very capable personal AI.

---

## 60. Definition of Success for v0.1

A.E.G.I.S. v0.1 should be considered successful when MAX can:

1. Hear the user.
2. Understand the request.
3. Remember relevant context.
4. Reason about the task.
5. Select an appropriate tool.
6. Check permission.
7. Execute the tool.
8. Observe the result.
9. Correct simple failures.
10. Explain what happened.
11. Maintain an audit record.

Example:

"MAX, open my development environment and tell me what's still pending in my CARLA project."

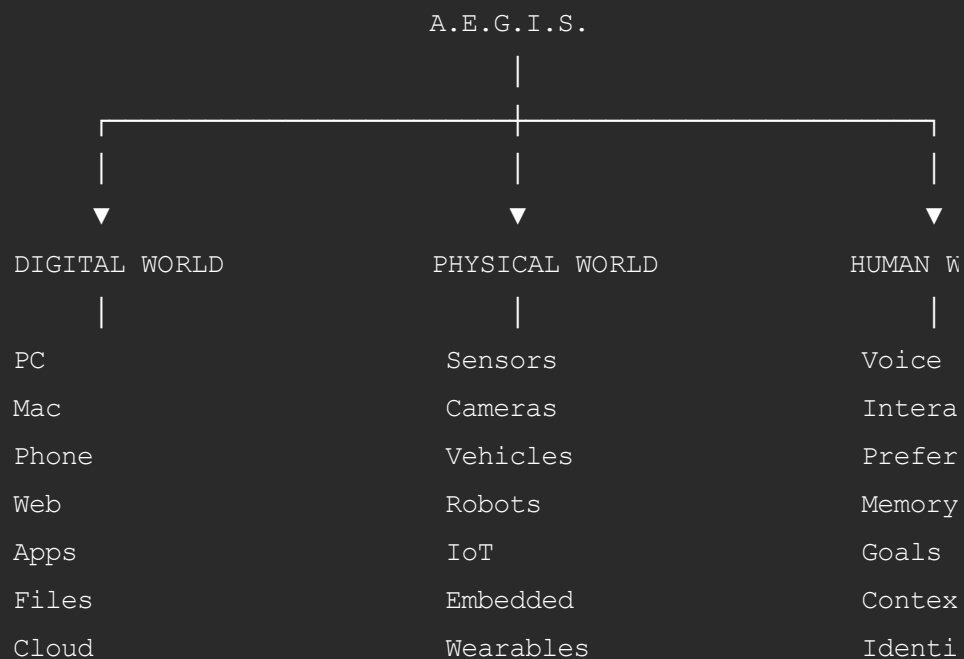
Expected pipeline:

Voice  
↓  
MAX  
↓  
Memory  
↓  
Project context  
↓  
Computer Agent  
↓  
IDE/files/Git  
↓  
Read status  
↓  
Reason  
↓  
Respond

That is already a legitimate personal AI system.

---

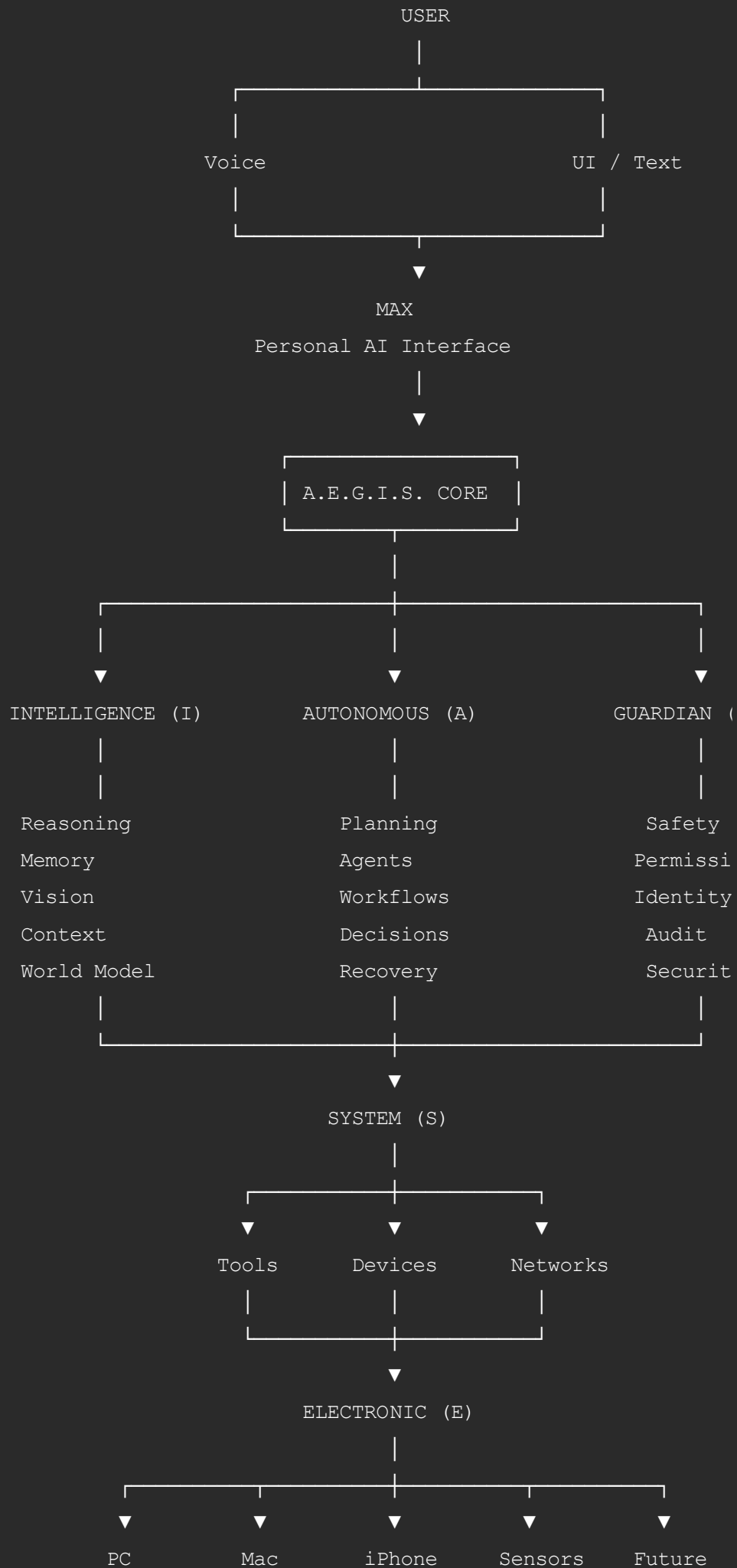
## 61. Long-Term A.E.G.I.S. Capability Map



Eventually all three become one contextual system.

---

## 62. Final Target Architecture





## 63. Final Project Definition

**A.E.G.I.S. — Autonomous Electronic Guardian Intelligence System** is a distributed personal artificial intelligence platform designed to provide persistent multimodal perception, reasoning, memory, planning, autonomous execution, and controlled interaction with digital and physical systems.

**MAX** is the human-facing intelligence and personality layer of A.E.G.I.S.

The system begins as a private software-based personal assistant and evolves progressively toward:

```
Personal Assistant
    ↓
Agentic Computer
    ↓
Persistent Personal Intelligence
    ↓
Multimodal World-Aware AI
    ↓
Multi-Device AI Platform
    ↓
Robotics / Vehicle Intelligence
    ↓
Physical Autonomous System
    ↓
Commercial A.E.G.I.S. Platform
```

The long-term architectural objective is therefore:

**Build one persistent intelligence that can understand the user, understand its environment, remember context, reason about goals, use tools, safely act across connected**

systems, and eventually inhabit different physical and digital platforms without changing its fundamental identity.

---

## 64. Recommended Immediate Development Sequence

The project should now move from concept to engineering in the following order:

```
PHASE 0
A.E.G.I.S. Architecture
      ↓
PHASE 1
Repository + Core Runtime
      ↓
PHASE 2
MAX Conversation Engine
      ↓
PHASE 3
Voice + Wake Word
      ↓
PHASE 4
Tool Registry
      ↓
PHASE 5
Computer Agent
      ↓
PHASE 6
Memory Engine
      ↓
PHASE 7
Guardian
      ↓
PHASE 8
Autonomous Agent
      ↓
PHASE 9
Vision
      ↓
PHASE 10
World Model
```



↓  
PHASE 11  
Sensor Fusion  
↓  
PHASE 12  
Multi-Device  
↓  
PHASE 13  
CARLA  
↓  
PHASE 14  
Physical Systems

## First engineering milestone

The first milestone should **not** be "build JARVIS."

It should be:

**Build a reliable A.E.G.I.S. core capable of receiving a natural-language request, reasoning about it, selecting a tool, passing through the Guardian, executing the action on the PC, observing the result, remembering the relevant outcome, and responding through MAX.**

Once that loop is real and reliable, everything else has somewhere sensible to attach.

The architecture is intentionally designed so that future sensor, mobile, robotics, vehicle, and cloud components extend the same platform rather than requiring a second project every time the laws of physics become inconvenient.

## Sources

The architecture above aligns with current technology directions including MCP's standardized tool/resource integration and its July 2026 security/routing updates, ROS 2's topic/service/action communication model for robotics, CARLA's sensor-streaming simulation model, and Apple's current on-device multimodal/tool-calling foundation-model APIs.

Android is also developing an agent-oriented AppFunctions layer that exposes application capabilities to authorized system agents, which is relevant to A.E.G.I.S.'s future cross-platform device architecture.

